COHOMOLOGY OF AN AFFINE SMOOTH CURVE WITH ROOTS OF UNITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete computation for all three parts; the references to earlier discussions are pointers to notation already introduced, not omissions of arguments, and the exact sequence together with the rank count is written out.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Cohomology of a smooth projective curve with roots of unity. Same chapter, Lemma lemma-cohomology-smooth-projective-curve. Supplies the rank count for the compactification and the divisibility discussion.
- Description of the first cohomology in terms of triples. Same chapter, Lemma lemma-describe-h1-mun and the discussion preceding it. Supplies the description of the first cohomology group and of the isomorphism classes of triples.

Source reference. Original remark 0H1F: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11250–11250. Navigation pointer invoked at proof line 11250; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03RQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11135–11135. Navigation pointer invoked at proof line 11234; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 040Q: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3057–3057. Navigation pointer invoked at proof line 11275; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be an affine smooth curve over an algebraically closed field k and $n \in k^*$. Let $X \subset \overline{X}$ be a smooth projective compactification (Varieties, Remark 0H1F). Let g be the genus of $\overline{X}$ and let r be the number of points of $\overline{X} \setminus X$. Then*

(1) $H^0_{étale}(X, \mu_n) = \mu_n(k)$;
(2) $H^1_{étale}(X, \mu_n) \cong (\mathbf{Z}/n\mathbf{Z})^{2g+r-1}$, *and*
(3) $H^q_{étale}(X, \mu_n) = 0$ *for all* $q \geq 2$.

Proof. Write $X = \overline{X} - \{x_1, \ldots, x_r\}$. Then $\mathrm{Pic}(X) = \mathrm{Pic}(\overline{X})/R$, where R is the subgroup generated by $\mathcal{O}_{\overline{X}}(x_i)$, $1 \leq i \leq r$. Since $r \geq 1$, we see that $\mathrm{Pic}^0(\overline{X}) \to \mathrm{Pic}(X)$ is surjective, hence $\mathrm{Pic}(X)$ is divisible (see discussion in proof of Lemma 03RQ). Applying the Kummer sequence, we get (1) and (3). For (2) we use

$$\begin{aligned} H^1_{étale}(X, \mu_n) &= \{(\mathcal{L}, \alpha)\}/\cong \\ &= (\{(\bar{\mathcal{L}}, D, \bar{\alpha})\}/\cong)/\tilde{R} \end{aligned}$$

For the first equality sign and notation see Lemma 040Q. The triples $(\bar{\mathcal{L}}, D, \bar{\alpha})$ in the second line consist of an invertible module $\bar{\mathcal{L}}$ on $\overline{X}$ of degree 0, a divisor D on $\overline{X}$ of degree 0 supported on $\{x_1, \ldots, x_r\}$ and an isomorphism $\bar{\alpha} : \bar{\mathcal{L}}^{\otimes n} \cong \mathcal{O}_{\overline{X}}(D)$. Isomorphisms of triples are defined as in the discussion preceding Lemma 040Q. Further $\tilde{R}$ is the subgroup generated by triples of the form $(\mathcal{O}_{\overline{X}}(D'), nD', 1^{\otimes n})$ where D' is supported on $\{x_1, \ldots, x_r\}$ and has degree 0. The discussion above shows we have the second equality sign via the map sending the triple $(\bar{\mathcal{L}},\ D,\ \bar{\alpha})$ to the pair $(\bar{\mathcal{L}}|_X, \bar{\alpha}|_X)$. Thus, we get an exact sequence

$$0 \longrightarrow H^1_{étale}(\overline{X}, \mu_n) \longrightarrow H^1_{étale}(X, \mu_n) \longrightarrow \bigoplus_{i=1}^{r} \mathbf{Z}/n\mathbf{Z} \xrightarrow{\Sigma} \mathbf{Z}/n\mathbf{Z} \longrightarrow 0$$

where the middle map sends the class of a triple $(\bar{\mathcal{L}}, D, \bar{\alpha})$ with $D = \sum_{i=1}^r a_i(x_i)$ to the r-tuple $(a_i)_{i=1}^r$. It now suffices to use Lemma 03RQ to count ranks. □